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COURSE: ITA0609-MACHINE LEARNING

CO4 - AT2 - INDUSTRY PROBLEM SOLVING

1. A telecommunications company wants to recommend suitable data plans to customers based on their usage patterns. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity between users is measured, how recommendations are generated, and how the system adapts to changing user behavior over time.

2. An energy management company aims to predict electricity consumption based on factors such as time, weather, and appliance usage. Propose a solution using Locally Weighted Regression or RBF networks. Explain how local learning captures variations in consumption patterns and how the model handles noisy and seasonal data.

Answers:

1. Data Plan Recommendation Using K-Nearest Neighbour (KNN)

Industry Problem

A telecommunications company has many customers with different data usage patterns. The company wants to automatically recommend the most suitable data plan to each customer.

1. Problem Identification

The system needs to:

- Analyze customer usage.
- Find customers with similar usage patterns.
- Recommend a suitable data plan.
- Adapt when the customer's usage changes.

2. Input Data

The system can use:

- Monthly data usage

- Call duration
- SMS usage
- Video streaming hours
- Social media usage
- Current plan

The target/class can be:

- Basic Plan
- Standard Plan
- Premium Plan

3. Similarity Measurement

KNN finds customers who are most similar to the new customer.

Euclidean distance is used:

$$d = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]}$$

A smaller distance means the customers are more similar.

Example:

Customer A: Data = 20 GB, Calls = 300 min

Customer B: Data = 25 GB, Calls = 320 min

$$\begin{aligned} d &= \sqrt{[(20-25)^2 + (300-320)^2]} \\ &= \sqrt{25 + 400} \\ &= 20.62 \end{aligned}$$

Therefore, they have relatively similar usage patterns.

4. Recommendation Process

Suppose $K = 5$.

The system finds the 5 customers most similar to the new customer.

Example:

Customer 1 → Standard

Customer 2 → Standard

Customer 3 → Premium

Customer 4 → Standard

Customer 5 → Standard

Standard = 4 votes

Premium = 1 vote

Recommended Plan = Standard Plan

5. How the System Adapts

Customer behavior changes over time. Therefore, the company continuously collects new information such as:

- Recent data consumption
- Changes in calling habits
- Streaming usage
- Weekend/weekday usage
- Current plan usage

Old usage data can be given less importance, while recent usage can be given more importance.

Example:

Old behavior: 10 GB/month → Basic Plan

New behavior: 50 GB/month → Premium Plan

The system detects the new pattern and recommends the Premium Plan.

6. Industry Solution

Customer Usage Data → Data Preprocessing → KNN → Find Similar Customers → Majority Voting → Data Plan Recommendation → Update with New Usage Data

7. Benefits

- Personalized recommendations.
- Simple and easy to implement.

- No complex training process.
- Adapts to changing customer behavior.
- Can improve customer satisfaction.
- Can help telecom companies increase plan upgrades.

Final Solution:

KNN is suitable because customers with similar usage patterns are likely to prefer similar data plans.

The system measures similarity using Euclidean distance, finds the K nearest customers, and recommends the plan with the highest number of votes. New customer usage data can continuously be added so that recommendations change according to recent behavior.

2. Electricity Consumption Prediction Using Locally Weighted Regression

Industry Problem

An energy management company wants to predict electricity consumption using:

- Time
- Weather
- Temperature
- Appliance usage
- Previous electricity consumption
- Day of week
- Season

1. Problem Identification

A normal regression model may assume that the relationship between input and electricity consumption is the same everywhere.

But in reality:

- Morning consumption is different from afternoon.
- Summer consumption is different from winter.
- Weekday consumption is different from weekend.
- Hot weather increases AC usage.

Therefore, a local learning method such as Locally Weighted Regression (LWR) can be used.

2. Input Data

Example:

Time	Temperature	AC Usage	Appliances	Consumption
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6 AM	25°C	1 hr	3	2.5 kWh
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9 AM	28°C	2 hrs	5	4.0 kWh
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1 PM	34°C	5 hrs	8	7.5 kWh
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6 PM	32°C	4 hrs	7	6.8 kWh
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10 PM	29°C	2 hrs	4	3.5 kWh
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3. Local Learning

LWR gives higher importance to data points that are close to the new input.

The weight can be calculated using:

$$w_i = e^{[-(x_i - x)^2 / (2\tau^2)]}$$

where:

- x_i = existing data point
- x = new input
- τ = bandwidth
- w_i = weight

A nearby data point gets a higher weight.

A far-away data point gets a lower weight.

4. Prediction

For a new customer:

Time = 1 PM

Temperature = 35°C

AC usage = 5 hours

The system searches for historical situations similar to these conditions.

Example:

1 PM, 34°C → 7.5 kWh → High weight

2 PM, 35°C → 8.0 kWh → High weight

12 PM, 33°C → 7.2 kWh → High weight

8 AM, 27°C → 4.0 kWh → Low weight

10 PM, 29°C → 3.5 kWh → Low weight

The nearby observations have higher influence.

Predicted consumption may therefore be around 7.5–8 kWh.

5. Handling Seasonal Data

Electricity usage changes according to seasons.

Summer:

- High temperature
- High AC usage
- High electricity consumption

Winter:

- Lower temperature
- Lower AC usage
- Lower electricity consumption

LWR focuses on observations close to the current conditions, so summer observations can have greater influence when predicting summer consumption.

6. Handling Noisy Data

Real-world energy data may contain noise because of:

- Sudden appliance usage
- Power fluctuations
- Sensor errors
- Unexpected weather
- Temporary changes in customer behavior

LWR reduces the effect of distant or less relevant observations by assigning them smaller weights.

The model therefore focuses mainly on relevant local data.

7. Industry Solution

Energy Data → Preprocessing → Select Similar Local Data → Assign Weights → Locally Weighted Regression → Predict Consumption → Compare with Actual Consumption → Update Data

8. Benefits

- Captures local consumption patterns.
- Handles changing relationships.
- Useful for seasonal electricity demand.
- Gives more importance to recent/relevant conditions.
- Can handle nonlinear consumption patterns.
- Useful for energy planning and demand management.

Final Solution:

Locally Weighted Regression is suitable because electricity consumption varies according to time, weather, appliance usage, and season. Instead of treating all historical data equally, LWR assigns higher weights to observations similar to the current situation. This allows the system to capture local variations and reduce the influence of irrelevant and noisy data.